

Renewable Energy Output Prediction — Case-Based Scenario

1. Original Problem Statement

Predict renewable energy output (solar and wind) using historical weather and plant operating data, comparing XGBoost and LightGBM regression models.

2. Case-Based Scenario

Background

GreenGrid Renewables, a mid-sized power distribution company operating in India, manages a portfolio of **10 renewable energy plants** — 5 solar farms (Rajasthan, Gujarat, Tamil Nadu, Karnataka, Madhya Pradesh) and 5 wind farms (Tamil Nadu, Gujarat, Karnataka, Andhra Pradesh, Maharashtra) — feeding electricity into the regional grid operated by the state **Load Dispatch Centre (LDC)**.

The Business Problem

Every day, GreenGrid’s grid-scheduling team must submit a **day-ahead generation forecast** to the LDC, declaring how much energy each plant will produce in every hour of the next day. This forecast matters because:

- **Under-forecasting** means GreenGrid loses revenue — it could have sold more power but didn’t commit to it.
- **Over-forecasting** triggers **grid-balancing penalties** — GreenGrid must pay for the shortfall when it fails to deliver the promised power, since the LDC has to activate expensive backup (usually thermal) power on short notice to fill the gap.
- Because solar output depends on sunlight/cloud cover and wind output depends on wind speed — both highly weather-dependent and variable hour to hour — manual or simple average-based forecasting has historically been inaccurate, costing the company an estimated **8–12% of potential monthly revenue** in penalties and missed sales.

The Ask

GreenGrid’s operations head has asked the data science team to build a **machine learning forecasting system** that:

1. Uses historical hourly weather readings (temperature, humidity, cloud cover, wind speed, precipitation, pressure, solar irradiance) and plant metadata (capacity, panel efficiency, turbine hub height, site, climate zone) to **predict hourly energy output (kWh)** for each plant.
2. Is accurate enough to reduce forecasting error to a level where grid penalties become negligible.
3. Can be explained to non-technical stakeholders — which weather factors drive output the most, and at which times of day/year output is highest or lowest — so the operations team can plan maintenance windows and energy trading strategy around it.
4. Compares **two candidate algorithms** — **XGBoost and LightGBM** — so the team can pick the one that gives the best accuracy/speed trade-off for daily retraining on an ever-growing dataset (the plants have now been running for 3 years, generating over 260,000 hourly readings).

Success Criteria (translated into ML metrics)

Business need	Technical metric
Forecast should be “close enough” on average	Low MAE (Mean Absolute Error) in kWh
Large forecast misses are especially costly (bigger penalties)	Low RMSE (penalizes big errors more)
Forecast should explain most of the variation in real output	High R^2
Team must trust <i>why</i> the model predicts what it predicts	Feature importance analysis
Model must generalize to <i>future</i> , unseen days — not just historical days it memorized	Time-based train/test split (train on 2022–2023, test on unseen 2024)

Deliverables Expected From the Data Science Team

1. **Exploratory Data Analysis (EDA)** — understand the data’s structure, quality, distributions, and relationships before modeling.
2. **Algorithm demonstration & implementation** — train, evaluate, and compare XGBoost and LightGBM models, and identify the key drivers of energy output for each source type.
3. A recommendation on which model to deploy into GreenGrid’s daily forecasting pipeline.